

Activity05

Ackaraphon Sahawiriya

6733291621

main.tf file

```
y05 > main.tf
terraform {
  required_providers {
    docker = {
      source  = "kreuzwerker/docker"
      version = "~> 2.21.0"
    }
  }
}

provider "docker" {}

# =====
# Docker Network
# =====

resource "docker_network" "todo_net" {
  name = "todo-net"
}

# =====
# Docker Images
# =====

# Redis
resource "docker_image" "redis" {
  name          = "redis:6-alpine"
  keep_locally = true
}

# Todo Service
resource "docker_image" "todo" {
  name          = "natawut/todo-service:release-3"
  keep_locally = true
}

# Todo Notification Service
resource "docker_image" "todo_notification" {
  name          = "activity03-todo-notification-service:latest"
  keep_locally = true
}

# =====
# Redis Container
# =====

resource "docker_container" "redis" {
  name = "redis"
  image = docker_image.redis.image_id

  networks_advanced {
    name = docker_network.todo_net.name
  }

  ports {
    internal = 6379
    external = 6379
  }
}

# =====
# Todo Notification Service
# =====

resource "docker_container" "todo_notification" {
  name = "todo-notification-service"
  image = docker_image.todo_notification.image_id

  networks_advanced {
    name = docker_network.todo_net.name
  }
}
```

```

ports {
  internal = 9000
  external = 9000
}
}

# =====
# Todo Service
# =====

resource "docker_container" "todo" {
  name = "todo-service"
  image = docker_image.todo.image_id

  networks_advanced {
    name = docker_network.todo_net.name
  }

  env = [
    "REDIS_HOST=redis",
    "NOTIFICATION_HOST=todo-notification-service",
    "NOTIFICATION_PORT=9000"
  ]

  ports {
    internal = 8000
    external = 8000
  }

  depends_on = [
    docker_container.redis,
    docker_container.todo_notification
  ]
}

```

Screenshot of running terraform in each step

Step 1: Terraform Initialization

```

ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ terraform init
Initializing the backend...

```

Initializing provider plugins...

- Finding kreuzwerker/docker versions matching "~> 2.21.0"...
- Installing kreuzwerker/docker v2.21.0...
- Installed kreuzwerker/docker v2.21.0 (self-signed, key ID BD080C4571C6104C)

Partner and community providers are signed by their developers.

If you'd like to know more about provider signing, you can read about it here:

<https://developer.hashicorp.com/terraform/cli/plugins/signing>

Terraform has created a lock file `.terraform.lock.hcl` to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands

Step 2: Terraform Configuration Validation

```

ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ terraform validate
Success! The configuration is valid.

```

Step 3: Terraform Plan

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# docker_container.redis will be created
+ resource "docker_container" "redis" {
  + attach      = false
  # docker_network.todo_net will be created
  + resource "docker_network" "todo_net" {
    + driver      = (known after apply)
    + id          = (known after apply)
    + internal    = (known after apply)
    + ipam_driver = "default"
    + name        = "todo-net"
    + options     = (known after apply)
    + scope       = (known after apply)

    + ipam_config (known after apply)
  }
}

Plan: 5 to add, 0 to change, 0 to destroy.
```

Step 4: Terraform Apply

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# docker_container.redis will be created
Enter a value: yes

docker_image.redis: Creating...
docker_image.todo: Creating...
docker_network.todo_net: Creating...
docker_image.redis: Creation complete after 0s [id=sha256:230747c6e7997406741497c085ccbfd1bdddff5c9c478f1ad4c3d7bfb5caedredis:6-alpine]
docker_network.todo_net: Creation complete after 2s [id=12666bdfc2ca744aa8ffc8e0fe91c3886175b15db060df20d6b7618f00267c5f]
docker_container.redis: Creating...
docker_container.redis: Creation complete after 1s [id=c69016b38795c4bcc3160f7d07bdf09fcf48d279ad2f8d15ba45348a85f17167]
docker_image.todo: Still creating... [00m10s elapsed]
docker_image.todo: Still creating... [00m22s elapsed]
docker_image.todo: Still creating... [00m32s elapsed]
docker_image.todo: Creation complete after 41s [id=sha256:ff25088c41c43b22a84e29ee2551a9193631d789ed2f56c305065645d024d9afnatawut/todo-service:release-3]
docker_container.todo: Creating...
docker_container.todo: Creation complete after 4s [id=b5fe72c6705332dcfa40974d547d0f8c9d3d0f88d2b035f2df468813fb3bc041]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.
```

Step 5: Verify Running Containers

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b5fe72c67053	ff25088c41c4	"/bin/sh -c 'python ..."	About a minute ago	Up About a minute	0.0.0.0:8000->8000/tcp	todo-service
c69016b38795	230747c6e799	"docker-entrypoint.s..."	About a minute ago	Up About a minute	0.0.0.0:6379->6379/tcp	redis

Step 6: Verify Docker Network

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ docker network inspect todo-net
[
  {
    "Name": "todo-net",
    "Id": "12666bdf2ca744aa8ffc8e0fe91c3886175b15db060df20d6b7618f00267c5f",
    "Created": "2026-09-11T16:54:15.029523253Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {
        "com.docker.network.enable_ipv4": "true"
      }
    },
    "Labels": {},
    "Containers": {
      "b5fe72c6705332dcfa40974d547d0f8c9d3d0f88d2b035f2df468813fb3bc041": {
        "Name": "todo-service",
        "EndpointID": "e307eae1b84815c365ebd545d8fcd0731a33ea2f9608277f8d594980393ea623",
        "MacAddress": "3a:4f:36:be:ba:b0",
        "IPv4Address": "172.20.0.3/16",
        "IPv6Address": ""
      },
      "c69016b38795c4bcc3160f7d07bdf09fcf48d279ad2f8d15ba45348a85f17167": {
        "Name": "redis",
        "EndpointID": "ede29c92031858e5d90a462279a0a7a26f3f23e24ed4b7c56aa8d6dd10d01511",
        "MacAddress": "ca:02:41:a3:e5:19",
        "IPv4Address": "172.20.0.2/16",
        "IPv6Address": ""
      }
    }
  }
]
```

Step 7: Verify Todo Service and Redis

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ docker logs todo-service
Connecting to Redis(redis, 6379)
Init Todo Service version 2.0
INFO: Started server process [7]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.20.0.1:39190 - "GET / HTTP/1.1" 200 OK

ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ docker logs redis
1:C 11 Sep 2026 16:54:18.276 # 0000000000000000 Redis is starting 000000000000
1:C 11 Sep 2026 16:54:18.276 # Redis version=6.2.24, bits=64, commit=00000000, modified=0, pid=1, just started
1:C 11 Sep 2026 16:54:18.276 # Warning: no config file specified, using the default config. In order to specify a config file use redis-server /path/to/redis.conf
1:M 11 Sep 2026 16:54:18.276 * monotonic clock: POSIX clock_gettime
1:M 11 Sep 2026 16:54:18.277 * Running mode=standalone, port=6379.
1:M 11 Sep 2026 16:54:18.278 # Server initialized
1:M 11 Sep 2026 16:54:18.279 * Ready to accept connections
```

Step 8: Check To-do Item

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ docker logs --tail 100 todo-service
Connecting to Redis(redis, 6379)
Init Todo Service version 2.0
INFO: Started server process [7]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.20.0.1:58072 - "GET / HTTP/1.1" 200 OK
{'id': 0, 'title': 'grader', 'timestamp': 1789148612}
INFO: 172.20.0.1:58074 - "POST / HTTP/1.1" 201 Created
INFO: 172.20.0.1:58090 - "GET / HTTP/1.1" 200 OK
```

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ curl http://localhost:8000
[{"id":0,"title":"grader","detail":{"check time 2026-09-12 00:43:32.049076096 +0700 +07 m=+8.670732190","completed":true,"duedate":"2026-09-12T00:43:32.049076+07:00","tags":[]}]ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$
```

```
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ chmod +x activity5-CEDT_linux_amd64
ackaraphon@DESKTOP-3ETTL2M:/mnt/d/CEDT/CEDTY3S1/SDS/Assignment/Activity05$ ./activity5-CEDT_linux_amd64
2026/09/12 00:43:23 Current Timestamp: 2026-09-12T00:43:23+07:00
👉 Enter [REQUIRED] Terraform file path (.tf) (leave blank for 'main.tf'):
2026/09/12 00:43:24 ✅ / Terraform file path is exist.
2026/09/12 00:43:24 ✅ / Terraform is installed.
2026/09/12 00:43:27 ✅ / Terraform is initialized.
2026/09/12 00:43:31 ✅ / Terraform plan is generated.
2026/09/12 00:43:32 ✅ / Container todo-service exists.
2026/09/12 00:43:32 ✅ / Container redis exists.
2026/09/12 00:43:32 ✅ / All specified containers are running.
2026/09/12 00:43:32 ✅ / URL http://localhost:8000 returns 200.
2026/09/12 00:43:32 ✅ / Todo is up and running at http://localhost:8000
2026/09/12 00:43:32 ✅ / POST request successful
2026/09/12 00:43:32 ✅ / POST request to http://localhost:8000 was successful.
2026/09/12 00:43:32 ✅ / GET request shows result from previous POST request to http://localhost:8000.
2026/09/12 00:43:32 Result: true
2026/09/12 00:43:32 🎉 Looks good! Please enter your StudentID and Full name below
👉 StudentID: 6733291621
👉 Full Name (TH): อัครพล สะหวังรัมย์
2026/09/12 00:44:43 🍀 Successfully submitted. Your lucky number is 21156991647900806
2026/09/12 00:44:43 🎉🎉🎉 Congratulations! You have completed the activity 🎉🎉🎉
2026/09/12 00:44:43 ⚠️ Don't forget! you still need to submit your assignment via MyCourseVille ⚠️
```